

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – CODING CHALLENGE

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 3 – Coding Challenge
Weightage:	30% of CIA		
CO5 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	Vivetha B	Reg. No.:	192424170
Section / Batch:	B. Tech (AI & DS)	Date:	14.07.2026

Code of Conduct

I Vivetha B 192424170 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Vivetha

Instructions

- Develop a complete and modular Python program that meets all the functional requirements specified in the coding challenge using appropriate algorithms, data structures, and programming practices.
- Test the program using multiple valid and invalid test cases, and clearly display the input, processing, and output to verify the correctness of the implementation.
- Follow standard coding practices by using meaningful variable names, proper indentation, comments, exception handling (where applicable), and upload the source code, sample outputs, and README file to the designated GitHub repository.
- Duration: 40 minutes.

Section / Topic	Focus	Q Nos
Regular Expression Pattern Matching	Regular Expressions, Basic Regular Expression Patterns	1
Finite State Automata Implementation	Finite State Automata, Models and Algorithms	2
Advanced Regular Expression Search	Regular Expressions, Basic Regular Expression Patterns	3

Section 1: Intelligent Email and Password Validator using Regular Expressions

1. Problem Statement:

A software company is developing a user registration system. Before creating an account, the system must validate user credentials based on predefined security policies.

Develop a Python application that validates:

Email Address

Password

Mobile Number

using Regular Expressions.

Validation Rules

Email

Must start with an alphabet.

May contain letters, digits, '.', '_' before '@'.

Domain should contain only alphabets.

Valid extensions: .com, .org, .edu, .net, .in

Password

Minimum 8 characters

At least one uppercase letter

At least one lowercase letter

At least one digit

At least one special character (@, #, \$, %, &, !)

Mobile Number

Must contain exactly 10 digits.

First digit should be between 6–9.

Expected Output

Display:

Valid Email / Invalid Email

Strong Password / Weak Password

Valid Mobile Number / Invalid Mobile Number

ANSWER:-

```
import re
email_pattern = r'^[A-Za-z][A-Za-z0-9._]*@[A-Za-z]+\.(com|org|edu|net|in)$'
password_pattern = r'^(?=.*[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[@#$$%&!]).{8,}$'
mobile_pattern = r'^[6-9]\d{9}$'
print("VALID INPUT")
email = "abc_123@gmail.com"
password = "Abc@1234"
mobile = "9876543210"
print("Email:", email)
if re.fullmatch(email_pattern, email):
    print("Valid Email")
else:
    print("Invalid Email")
print("Password:", password)
if re.fullmatch(password_pattern, password):
    print("Strong Password")
else:
    print("Weak Password")
print("Mobile Number:", mobile)
if re.fullmatch(mobile_pattern, mobile):
    print("Valid Mobile Number")
```

```

else:
    print("Invalid Mobile Number")
print("\nINVALID INPUT")
email = "1abc@gmail"
password = "abc123"
mobile = "5123456789"
print("Email:", email)
if re.fullmatch(email_pattern, email):
    print("Valid Email")
else:
    print("Invalid Email")
print("Password:", password)
if re.fullmatch(password_pattern, password):
    print("Strong Password")
else:
    print("Weak Password")
print("Mobile Number:", mobile)
if re.fullmatch(mobile_pattern, mobile):
    print("Valid Mobile Number")
else:
    print("Invalid Mobile Number")

```

```

==== RESTART: C:/Users/bvbi/AppData/Local/Programs/Python/Python313/zvsv.py ===
VALID INPUT
Email: abc_123@gmail.com
Valid Email
Password: Abc@1234
Strong Password
Mobile Number: 9876543210
Valid Mobile Number

INVALID INPUT
Email: 1abc@gmail
Invalid Email
Password: abc123
Weak Password
Mobile Number: 5123456789
Invalid Mobile Number

```

Section 2: Design a Finite State Automata (FSA) Simulator

2. Problem Statement

Develop a Python-based Deterministic Finite Automata (DFA) simulator.

The program should:

- Accept the DFA description
 - States
 - Input alphabet
 - Transition table
 - Initial state
 - Final state(s)
- Accept multiple input strings.
- Simulate state transitions.
- Display whether the string is Accepted or Rejected.

Example

Language:
Strings ending with "ab"

Input

abaab

Output

Transition Path:
 $q_0 \rightarrow q_1 \rightarrow q_0 \rightarrow q_1 \rightarrow q_2$
Accepted

ANSWER:-

```
states = ['q0', 'q1', 'q2']
```

```
alphabet = ['a', 'b']
```

```
transition = {  
    ('q0', 'a'): 'q1',  
    ('q0', 'b'): 'q0',  
    ('q1', 'a'): 'q1',  
    ('q1', 'b'): 'q2',  
    ('q2', 'a'): 'q1',  
    ('q2', 'b'): 'q0'  
}
```

```
initial_state = 'q0'
```

```
final_states = ['q2']
```

```
print("VALID INPUT")  
string = "abaab"
```

```
current_state = initial_state  
path = [current_state]
```

```
for symbol in string:  
    current_state = transition[(current_state, symbol)]  
    path.append(current_state)
```

```
print("Input String:", string)  
print("Transition Path:", " → ".join(path))
```

```
if current_state in final_states:  
    print("Accepted")  
else:  
    print("Rejected")
```

```
print("\nINVALID INPUT")  
string = "ababa"
```

```
current_state = initial_state  
path = [current_state]
```

```
for symbol in string:  
    current_state = transition[(current_state, symbol)]  
    path.append(current_state)
```

```
print("Input String:", string)
```

```
print("Transition Path:", " → ".join(path))
```

```
if current_state in final_states:  
    print("Accepted")  
else:  
    print("Rejected")
```

```
= RESTART: C:/Users/bvbvi/AppData/Local/Programs/  
on313/zvsv.py
```

```
VALID INPUT
```

```
Input String: abaab
```

```
Transition Path: q0 → q1 → q2 → q1 → q1 → q2
```

```
Accepted
```

```
INVALID INPUT
```

```
Input String: ababa
```

```
Transition Path: q0 → q1 → q2 → q1 → q2 → q1
```

```
Rejected
```

```
|
```

Section 3: Smart Pattern Matching Engine using Regular Expressions

3. Problem Statement

Develop a Python-based text search engine that performs pattern matching using Regular Expressions.

The system should support:

- Word Search
- Prefix Search
- Suffix Search
- Date Search
- Phone Number Search
- Hashtag Search
- Mention (@username) Search

Example Input

Meeting on 12/09/2026
Call 9876543210
#NLP
@OpenAI
natural language processing

User Menu

- 1.Search Date
- 2.Search Phone Number
- 3.Search Hashtag
- 4.Search Mention
- 5.Search Prefix
- 6.Search Suffix

Expected Output

Display all matching patterns based on user selection.

ANSWER:-

```
import re

text = """
Meeting on 12/09/2026
Call 9876543210
#NLP
@OpenAI
natural language processing
machine learning
"""

print("Sample Text:")
print(text)

print("Menu")
print("1. Search Date")
print("2. Search Phone Number")
print("3. Search Hashtag")
print("4. Search Mention")
print("5. Search Prefix")
print("6. Search Suffix")
print("7. Search Word")

choice = int(input("Enter your choice: "))

if choice == 1:
    result = re.findall(r'\b\d{2}/\d{2}/\d{4}\b', text)
    print("Date Found:", result)

elif choice == 2:
    result = re.findall(r'\b[6-9]\d{9}\b', text)
    print("Phone Number Found:", result)

elif choice == 3:
    result = re.findall(r'#\w+', text)
    print("Hashtag Found:", result)

elif choice == 4:
    result = re.findall(r'@\w+', text)
    print("Mention Found:", result)

elif choice == 5:
    prefix = input("Enter Prefix: ")
    result = re.findall(r'\b' + prefix + r'\w*', text)
    print("Matching Words:", result)

elif choice == 6:
    suffix = input("Enter Suffix: ")
    result = re.findall(r'\b\w*' + suffix + r'\b', text)
    print("Matching Words:", result)

elif choice == 7:
    word = input("Enter Word: ")
    result = re.findall(r'\b' + word + r'\b', text)
    print("Word Found:", result)

else:
    print("Invalid Choice")
```

```
= RESTART: C:/Users/bvbvi/AppData/Local/Programs  
on313/zvsv.py  
Sample Text:
```

```
Meeting on 12/09/2026  
Call 9876543210  
#NLP  
@OpenAI  
natural language processing  
machine learning
```

```
Menu
```

1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Search Word

```
Enter your choice: 2
```

```
Phone Number Found: ['9876543210']
```

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases

Q. No. / Criteria	Problem Understanding	Algorithm	Code Implementation	Competitive Performance	Test Case Handling	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____